

A strict pointwise tensor-train approximation bound by finite singular-subspace tie handling

Recovered qualified manuscript for TR-04. Not a uniform-factor improvement.

Abstract

For prescribed tensor-train rank bounds, a deterministic algorithm in the exact arithmetic/SVD model returns an approximation with squared error strictly smaller than $(d - 1)$ times the optimal squared error whenever that optimum is positive, and reconstructs exactly at zero optimum. It performs at most n_1 tensor-train SVD completions. The only modification to the standard construction is to test a finite collection of equally optimal first-cut singular subspaces when a positive singular value straddles the truncation boundary. This addresses the strict pointwise inequality displayed in TR-04. It does not produce a uniform constant $c < d - 1$: an explicit fixed-format family shows that its error ratios can approach $d - 1$ already when $d = 3$.

1 The target and the distinction between two guarantees

Let $d \geq 3$, let $A \in \mathbb{R}^{n_1 \times \cdots \times n_d}$ be a dense tensor, and let $r_1, \dots, r_{d-1}$ be positive integers. Denote by S_r the set of tensors whose unfolding at every cut $1, \dots, j \mid j + 1, \dots, d$ has rank at most r_j . Write

$$E_* = E_*(A, r) = \min_{Y \in S_r} \|A - Y\|_F^2.$$

This minimum exists: S_r is a closed intersection of determinantal sets, and a minimizing sequence may be confined to a bounded ball. The displayed strict inequality in the repository is

$$\|A - X\|_F^2 < (d - 1)E_* \quad (E_* > 0), \tag{1}$$

with exact reconstruction when $E_* = 0$, in the idealized arithmetic/SVD model [3, 2]. A guarantee of the form (1) for every input does not imply that the supremum of all error ratios is smaller than $d - 1$. We prove the former, and make no claim of the latter. In particular this note must not be presented as a smaller uniform worst-case constant.

2 A completion estimate

The standard TT-SVD guarantee [1, Theorem 2.2 and Corollary 2.4] will be used on the remaining cuts after a first projection.

Lemma 1. *Let $U \in \mathbb{R}^{n_1 \times k}$ have orthonormal columns, with $k \leq r_1$, and put $P = UU^T$. Compress the first mode to form $B = U^T A$. Treat its first two indices as a single index, so B is a tensor of order $d - 1$ and format $(kn_2) \times n_3 \times \cdots \times n_d$. Apply ordinary TT-SVD to this tensor with cut-rank bounds $r_2, \dots, r_{d-1}$, and lift the result through U to obtain X_U . Then $X_U \in S_r$, and for every $Y \in S_r$,*

$$\|A - X_U\|_F^2 \leq \|(I - P)A\|_F^2 + (d - 2)\|P(A - Y)\|_F^2. \tag{2}$$

Here and below a matrix acting on a tensor acts on its first mode.

Proof. The first cut of X_U has rank at most k . The remaining cuts have the specified rank bounds, which are preserved by lifting through U . The compressed tensor $U^T Y$ satisfies every remaining cut constraint: compression is left multiplication on each such unfolding and cannot increase its rank. The TT-SVD guarantee on the $d - 2$ remaining cuts is therefore

$$\|B - \hat{B}\|_F^2 \leq (d - 2)\|U^T(A - Y)\|_F^2.$$

For completeness, this usual guarantee follows by summing the squared orthogonal truncation residuals at successive cuts. Each residual is at most the best error at that cut; projecting any feasible comparison tensor through the previously computed orthogonal factors shows that each is at most the squared error of that comparison tensor. There are $d - 2$ cuts. This argument also gives zero residuals when the input is already feasible. Finally, the first-mode discarded component is orthogonal to the lifted completion residual, giving

$$\|A - X_U\|_F^2 = \|(I - P)A\|_F^2 + \|B - \hat{B}\|_F^2.$$

The identity $\|U^T(A - Y)\|_F = \|P(A - Y)\|_F$ completes the proof. $\square$

3 The finite algorithm

Let $A_{(1)}$ denote the $n_1 \times (n_2 \cdots n_d)$ first unfolding. If $A = 0$, return 0. Compute its SVD and let

$$\varepsilon^2 = \sum_{i > r_1} \sigma_i(A_{(1)})^2,$$

where singular values beyond the matrix dimensions are interpreted as zero. Construct a list $\mathcal{P}$ of first-mode projectors as follows.

If $\text{rank } A_{(1)} \leq r_1$, use just the projector onto the column space of $A_{(1)}$. Otherwise let $\tau = \sigma_{r_1}(A_{(1)}) > 0$. Let H be the span of left singular vectors with singular value strictly greater than τ , put $h = \dim H$, and let E be the left singular space for τ , of dimension t . Set $s = r_1 - h$, so $1 \leq s \leq t$. If $s = t$, use just the projector onto $H \oplus E$. If $s < t$, take any fixed orthonormal basis $u_0, \dots, u_{t-1}$ of E and form the t cyclic windows

$$W_j = \text{span}\{u_j, u_{j+1}, \dots, u_{j+s-1}\}, \quad 0 \leq j < t,$$

with subscripts modulo t . Use the t projectors

$$P_j = P_H + P_{W_j}. \tag{3}$$

Each projector in $\mathcal{P}$ is a best first-cut rank- r_1 projection (or an exact column-space projection when the rank is smaller). For each $P \in \mathcal{P}$, choose an orthonormal column matrix U with $P = UU^T$, form X_U by Lemma 1, and compute its actual Frobenius error. Return a candidate of minimum error, resolving any final tie deterministically.

There are at most n_1 candidates. SVDs, matrix products, tensor reshapings, and error computations have polynomial cost in the dense input size and the rank parameters. Thus this is a deterministic polynomial-time algorithm in the arithmetic/SVD model. Equality of singular values is tested exactly in that model. A floating-point implementation with a tolerance is an illustration, not a bit-complexity implementation of the exact tie test.

4 Proof of the strict guarantee

We first isolate the only equality case requiring attention.

Lemma 2. Suppose $\text{rank } A_{(1)} > r_1$, and let Y satisfy $\text{rank } Y_{(1)} \leq r_1$ and $\|A - Y\|_F^2 = \varepsilon^2$. With H and E defined above, the residual $R = A - Y$ obeys $P_H R = 0$. If $s = t$, it also obeys $(P_H + P_E)R = 0$.

Proof. Let S be the column space of $Y_{(1)}$. The orthogonal decomposition

$$\|A_{(1)} - Y_{(1)}\|_F^2 = \|(I - P_S)A_{(1)}\|_F^2 + \|P_S A_{(1)} - Y_{(1)}\|_F^2$$

shows that equality with the best rank- r_1 error forces $Y_{(1)} = P_S A_{(1)}$ and forces P_S to maximize the captured singular energy. Since $\tau > 0$, $\dim S = r_1$. The equality characterization of this variational maximum gives $S = H \oplus E_0$ for an s -dimensional subspace $E_0 \subset E$. One can see this directly by writing $\text{tr}(P_S A_{(1)} A_{(1)}^T)$ in an eigenbasis: all weights at eigenvalues strictly above τ^2 must be one, all below must be zero, and the remaining weight is s in the τ^2 eigenspace. Thus $H \subset S$, and also $E \subset S$ when $s = t$. The asserted residual orthogonality follows. $\square$

Theorem 3. The finite algorithm above returns $X \in S_r$ satisfying (1) whenever $E_* > 0$, and returns $X = A$ when $E_* = 0$.

Proof. Fix a best feasible $Y \in S_r$, and write $R = A - Y$. The first-cut lower bound is $\varepsilon^2 \leq E_*$. Every tested projector P has $\|(I - P)A\|_F^2 = \varepsilon^2$; Lemma 1 gives

$$\|A - X_U\|_F^2 \leq \varepsilon^2 + (d - 2)\|PR\|_F^2 \leq \varepsilon^2 + (d - 2)E_*. \quad (4)$$

If $\varepsilon^2 < E_*$, this is strictly smaller than $(d - 1)E_*$. In particular, this covers $\text{rank } A_{(1)} \leq r_1$ whenever $E_* > 0$.

It remains to consider $\varepsilon^2 = E_* > 0$. By Lemma 2, $P_H R = 0$. If $s = t$, the single tested projector annihilates R , so (4) bounds the error by $E_* < (d - 1)E_*$. If $0 < s < t$, the cyclic-window projectors satisfy the exact averaging identity

$$\frac{1}{t} \sum_{j=0}^{t-1} P_{W_j} = \frac{s}{t} P_E,$$

since each basis vector appears in exactly s windows. Hence

$$\frac{1}{t} \sum_{j=0}^{t-1} \|P_j R\|_F^2 = \frac{s}{t} \|P_E R\|_F^2 \leq \frac{s}{t} E_* < E_*.$$

At least one tested projector has this bound. Its completion satisfies

$$\|A - X_U\|_F^2 \leq \left(1 + (d - 2)\frac{s}{t}\right) E_* < (d - 1)E_*.$$

The returned minimum-error candidate is no worse.

Finally, if $E_* = 0$, then $A \in S_r$, $\varepsilon^2 = 0$, and (4) with $R = 0$ forces the returned error to be zero. $\square$

5 Why this is not a uniform constant improvement

This distinction is material to the scope of the result, not a numerical precision issue. Consider $d = 3$, format $3 \times 3 \times 2$, ranks $(2, 1)$, and the tensor

$$A = \alpha e_1 \otimes e_1 \otimes f_1 + \beta e_2 \otimes e_2 \otimes f_1 + \gamma e_3 \otimes e_3 \otimes f_2, \quad \alpha > \gamma \geq \beta > 0.$$

Its optimal error is $E_* = \gamma^2$. Indeed, a tensor feasible for ranks $(2, 1)$ is $M \otimes v$ with $\text{rank } M \leq 2$. For a unit vector $v = (c, s)$, the contracted matrix is $\text{diag}(\alpha c, \beta c, \gamma s)$. The maximum possible sum of its two largest squared singular values, also maximizing over $c^2 + s^2 = 1$, is

$$\max\{\alpha^2 + \beta^2, \alpha^2, \beta^2, \gamma^2\} = \alpha^2 + \beta^2.$$

This proves the claimed optimum by orthogonal projection.

When $\gamma > \beta$, the best first-cut rank-two subspace is uniquely $\text{span}(e_1, e_3)$. The remaining rank-one truncation retains only the α term. The algorithm's squared error is therefore $\beta^2 + \gamma^2$, and its ratio to optimum is

$$1 + (\beta/\gamma)^2 < 2.$$

Keeping $\alpha = 2$, $\beta = 1$ and sending $\gamma \downarrow 1$ shows that these ratios approach $2 = d - 1$ in a fixed format. At $\gamma = \beta$, an unfortunate ordinary TT-SVD tie choice can attain ratio exactly two; the finite tie handling removes that equality. No theorem here asserts a uniform ratio bounded by $2 - \delta$ for a fixed $\delta > 0$.

6 Reproducibility and scope

The accompanying `verify_tr04.py` checks the projector averaging identity exactly, checks the fixed-format family and rotated tie bases, and tests the numerical implementation for rank feasibility and exact reconstruction up to floating-point error. The exact mathematical algorithm and its guarantee are Theorem 3; a tolerance-based SVD is not claimed to detect exact ties in finite precision. No ranks are enlarged. No oracle for E_* or for an optimal Y is used: those objects appear only in the proof. The assertion addressed is the strict pointwise target, not a stronger uniform-factor or bit-complexity formulation.

References

- [1] I. V. Oseledets, *Tensor-Train Decomposition*, SIAM Journal on Scientific Computing 33 (2011), 2295–2317, Theorem 2.2 and Corollary 2.4. doi:10.1137/090752286.
- [2] N. Amsel et al., *Linear Systems and Eigenvalue Problems: Open Questions from a Simons Workshop*, arXiv:2602.05394v3 (2026), Section 6.1, Problem 6.1.
- [3] OpenProblemsInNLA, *TR-04: Improve the worst-case approximation factor for pre-scribed tensor-train ranks*, canonical statement. <https://github.com/ajt60gaibb/OpenProblemsInNLA/tree/main/tensor-computations/TR-04>.